M T W T F S S
Page No.:
Date:
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Semi-Major Test-3 (Maths)

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Q.1 $\cancel{D_S = D_R} \Rightarrow \cancel{S_S \times 36 = S_R \times 55}$ - ① $\left(S = \frac{D}{t}\right)$
$\cancel{S_S = \frac{D}{t} \Rightarrow S_S = \frac{D}{36}}$

$AP_1 = 36, 72, 108, 144, 180$
$AP_2 = 45, 90, 135, 180$
$\therefore t = 3 \text{ min}$ (Option 3)

Q.2 $\dfrac{\frac{1}{\sqrt{3}}}{1-\frac{1}{3}} = \dfrac{\frac{1}{\cancel{\sqrt{3}}}}{\frac{2}{\sqrt{3}\times\cancel{\sqrt{3}}}} = \frac{\sqrt{3}}{2} = \sin 60°$
(2)

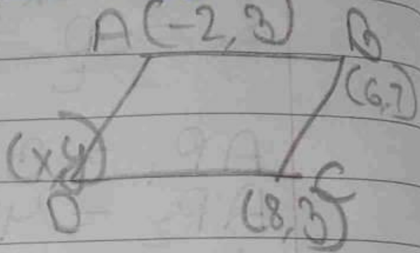
Q.3 $\frac{BP}{CP} = \frac{AP}{DP}$ and $\angle P = \angle P$
$\therefore \triangle BPA \sim \triangle CPD \Rightarrow \angle B = \angle C = 100°$
(Option 4)

Q.4 $\frac{AB}{QR} = \frac{BC}{PR} = \frac{AC}{QP} = \frac{3}{2}$
$\Rightarrow \frac{18}{QR} = \left(\frac{\cancel{15}\,5}{PR} = \frac{\cancel{3}}{2}\right) \Rightarrow PR = 10 \text{ cm}$ (1)

Q.5 $4+2+2a+b=0 \Rightarrow 2a+b=-6$ - ①
$9-3-3a+b=0 \Rightarrow -3a+b=-6$ - ②
$\Rightarrow a=0$

Page No.:
Date:
YOUVA

Ans ⇒ (4) 0.
Q.6 (4) 4 decimal places.
Q.7 $a - b = 2$
$a + b = 4$
$a = 3, b = 1$ (Option (3))



Q.8 $6 = 6 + x \Rightarrow x = 0$
$6 = 7 + y \Rightarrow y = -1$
Ans ⇒ (2) $(0, -1)$.

Q.9 (4) No solution.
Q.10 $\frac{2}{3} = \frac{-k-3}{k} \Rightarrow k = \frac{-9}{2}$ Option (1)
Q.11 $\frac{\cancel{3}}{3} \neq \frac{-\cancel{3}}{k} \Rightarrow k \neq -3$ (1)
Q.12 (1) 8.
Q.13 (1) similar.
Q.14 $LCM = x^2 y z^2$ (2)
Q.15 $\alpha + \beta = 1$ — (1)
$\alpha\beta = -12$ — (2)
$\Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta$ =
$\Rightarrow x^2 - x - 12 \Rightarrow \frac{x^2}{2} - \frac{x}{2} - 6$ (3)

Q.16 (3) Intersecting or coincident.
Q.17 (1) Parallel.
Q.18 (4) 16 : 81.

Q.19 (3) - Assertion (A) is true but Reason (R) is false.

Q.20 $\frac{1}{\cancel{5}} = \frac{3R-3}{\cancel{15}\,3} \Rightarrow R-1 = 1 \Rightarrow \underline{R=2}$

A ⇒ (1) Both A and R are true and R is the correct explanation of A.

Q.21 To prove ⇒ $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ is a composite number

Proof ⇒ ~~720 + 5 = 5~~

$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$= 720 + 5$

$= 5(145 + 1) = 5 \times 146$

Now, WKT, a number is a composite number if it has more than 2 factors or divisible by any other number except 1 and the number itself.

∴ Clearly, the given number has other factors.

∴ The number $(6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5)$ is a composite number.

Hence, ~~Proof~~ Proved.

Q.22 Given - Eq① $= 4x+6y = 7$
Eq② $= kx+18y = 21$
The equations have infinite solu-
-tions
To find - Value of 'k'
Sol - WKT, In the case of infinitely
many solutions,
$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$
Here, $a_1 = 4$, $a_2 = k$
$b_1 = 6$, $b_2 = 18$
$c_1 = -7$, $c_2 = -21$
$$\therefore \frac{4}{k} = \frac{6}{18} = \frac{-7}{-21}$$
$$\Rightarrow \frac{4}{k} = \frac{1}{3} \Rightarrow \boxed{k = 12}$$

Q.23 To find - Distance between origin $(0,0)$
and $(-6, 8)$
Sol, We know that -
Distance between a point and
origin is given by formula,
$$\Rightarrow \text{Distance} = \sqrt{(x_1)^2 + (y_1)^2}$$
Here, $x_1 = -6$ and $y_1 = 8$

$\therefore$ Distance $= \cancel{\sqrt{36} +} \sqrt{(-6)^2 + (8)^2}$

$= \sqrt{36 + 64} = \sqrt{100}$

$= 10$ units

Q.24 Given - $\sin 3A = \cos(A - 10^\circ)$

To find - Value of A

Sol- $\sin 3A = \cos(A - 10^\circ)$

Now, WKT -

$\Rightarrow \cos\theta = \sin(90 - \theta)$

Here, $\theta = A - 10^\circ$

$\therefore$ ~~$\cos(A - 10^\circ) = \sin(90 - 3A)$~~

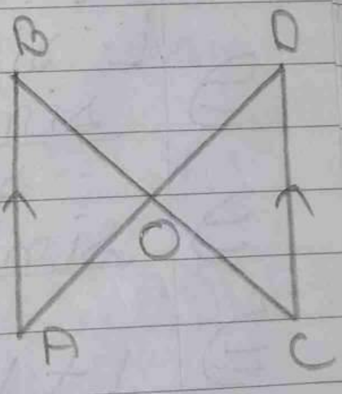
$\therefore \cos(A - 10^\circ) = \cancel{\cos} \sin(90^\circ - A + 10^\circ)$

$\Rightarrow \sin 3A = \sin(100^\circ - A)$ (Given)

$\Rightarrow 3A = 100^\circ - A$

$\Rightarrow 3A + A = 100^\circ \Rightarrow 4A = 100^\circ$

$\Rightarrow A = \frac{100^\circ}{4} = \boxed{25^\circ}$



Q.25 Given - AB || CD

To prove - $\Delta AOB \sim \Delta DOC$

Sol- We know that,

Alternate Interior Angles are equal

$\therefore \angle DAB = \angle ADC$ - ① and

$\angle ABC = \angle BCD$ - ②

Now, In ΔAOB and ΔDOC

$\Rightarrow \angle ABO = \angle DCO$ (Proved above)
$\Rightarrow \angle BAO = \angle CDO$ (Proved above)
$\Rightarrow \angle AOB = \angle DOC$ (Vertically Opp. Angle)
$\Rightarrow \Delta AOB \sim \Delta DOC$ (AAA Similarity)

Hence, Proved.

Q.27

Q.26 Taking LHS,

$$\Rightarrow \frac{\cos(90^\circ-\theta)}{1+\sin(90^\circ-\theta)} + \frac{1+\sin(90^\circ-\theta)}{\cos(90^\circ-\theta)}$$

$$\Rightarrow \frac{\sin\theta}{1+\sin(90^\circ-\theta)} + \frac{1+\sin(90^\circ-\theta)}{\sin\theta}$$

$[\because \cos(90^\circ-\theta) = \sin\theta]$

$$\Rightarrow \frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$$

$[\because \sin(90^\circ-\theta) = \cos\theta]$

$$\Rightarrow \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$$

$[(a+b)^2 = a^2+b^2+2ab]$

$$\Rightarrow \frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$\Rightarrow \frac{1+1+2\cos\theta}{\sin\theta(1+\cos\theta)} \quad [\because \sin^2\theta+\cos^2\theta = 1]$$

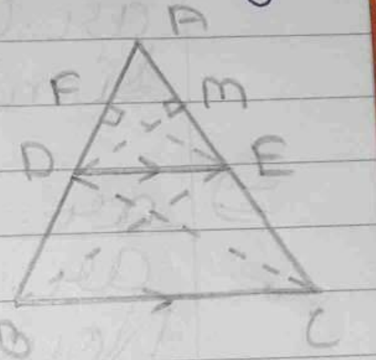
$$\Rightarrow \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)} = \frac{2}{\sin\theta}\left(\frac{\cancel{1+\cos\theta}}{\cancel{1+\cos\theta}}\right)$$

$$\Rightarrow 2\operatorname{cosec}\theta = RHS \quad \left[\because \frac{1}{\sin\theta} = \operatorname{cosec}\theta\right]$$

Hence, Proved.

Q.27 Basic Proportionality Theorem (BPT)

Basic Proportionality Theorem states that if a line segment parallel to one side of a triangle intersects the other two sides of the same triangle then the line segment divides the sides of the triangle in same ratio.



Proof -

Given - $DE \parallel BC$

To prove - $\frac{AD}{BD} = \frac{AE}{CE}$

~~Proof~~ Construction -

Join BE and CD. Draw $FE \perp AD$ and $DM \perp AE$

Proof - Area of $\Delta DBE = \frac{1}{2} \times$ base $\times$ height

$\Rightarrow ar.(\Delta DBE) = \frac{1}{2} \times BD \times FE$ - (1)

[$\because FE \perp AD \therefore FE$ is height of ΔDBE]

$\Rightarrow ar.(\Delta ADE) = \frac{1}{2} \times$ base $\times$ height

$\Rightarrow ar.(\Delta ADE) = \frac{1}{2} \times AD \times FE$ - (2)

from eq. 1 and 2

$$\Rightarrow \frac{ar.(\triangle ADE)}{ar.(\triangle DBE)} = \frac{\cancel{\frac{1}{2}} \times AD \times \cancel{EF}}{\cancel{\frac{1}{2}} \times BD \times \cancel{EF}}$$

$$\Rightarrow \frac{ar.(\triangle ADE)}{ar.(\triangle DBE)} = \frac{AD}{BD} \quad -③$$

Similarly,

$$\Rightarrow \frac{ar.(\triangle ADE)}{ar.(\triangle DCE)} = \cancel{\frac{AE}{CE}} \; \cancel{-④} \; \frac{AE}{CE} \quad -④$$

Now, We know that Triangles formed between same parallels have equal area

Here, $DE \parallel BC$

$\therefore$ ar.$(\triangle DBE)$ = ar.$(\triangle DCE)$

$$\Rightarrow \frac{1}{ar.(\triangle DBE)} = \frac{1}{ar.(\triangle DCE)} \quad \text{(Reciprocal)}$$

$$\Rightarrow \frac{ar.(\triangle ADE)}{ar.(\triangle DBE)} = \frac{ar.(\triangle ADE)}{ar.(\triangle DCE)}$$

[Multiply both sides by ar.$(\triangle ADE)$.

$$\Rightarrow \frac{AD}{BD} = \cancel{\frac{AE}{CE}}\frac{AE}{CE} \quad \text{(from ③ and ④)}$$

Hence, Proved.

Q.28 Let the fixed charges for 2 days be ₹x and additional charge for each day thereafter ~~by~~ be ₹y.

ATQ -

$\Rightarrow x + (6-2)y = ₹22 \Rightarrow x + 4y = 22$ — (1)

$\Rightarrow x + (4-2)y = ₹16 \Rightarrow x + 2y = 16$ — (2)

from (1) and (2)

$$\Rightarrow \begin{array}{r} \cancel{x} + 4y = 22 \\ -\cancel{x} \mp 2y = 16 \\ \hline 2y = 6 \end{array} \Rightarrow y = \frac{6}{2} = ₹3$$

$\Rightarrow y = ₹3$ — (3)

from (2) and (3)

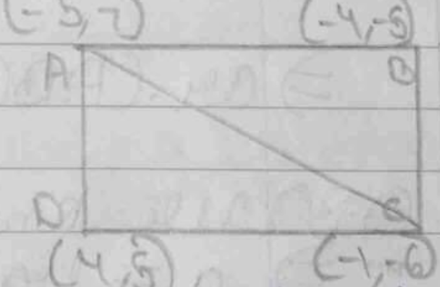
$\Rightarrow x + 2(3) = 16 \Rightarrow x + 6 = 16$

$\Rightarrow x = 16 - 6 = ₹10$

Ans → Fixed charges for 2 days = ₹10

Additional charge for each day thereafter = ₹3

Q.29 To find - ar. (ABCD)



Sol - WKT -

area of ABCD = area of $\triangle ABC$ + ar. ($\triangle ACD$) — (1)

Now, ar. ($\triangle$) = $\left| \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \right|$

M T W T F S S
Page No.:
Date:
YOUVA

$\therefore$ area of $\Delta ABC = \left| \frac{1}{2}[-5(1) - 4(-13) - 1(12)] \right|$

$\Rightarrow$ ar. $(\Delta ABC) = \left| \frac{1}{2}(-5 + 52 - 12) \right|$

$\Rightarrow$ ar. $(\Delta ABC) = \left| \frac{1}{2} \times 35 \right| = \frac{35}{2}$ unit^2 — (2)

ar. $(\Delta ACD) = \left| \frac{1}{2}[-5(-11) - 1(-2) + 4(13)] \right|$

$\Rightarrow$ ar. $(\Delta ACD) = \left| \frac{1}{2}(55 + 2 + 52) \right|$

$\Rightarrow$ ar. $(\Delta ACD) = \left| \frac{109}{2} \right| = \frac{109}{2}$ unit^2 — (3)

from (1), (2) and (3)

$\Rightarrow$ area of quad. ABCD = area of (ΔABC) + area of ΔACD

$\Rightarrow$ ar. $(ABCD) = \frac{35}{2} + \frac{109}{2}$

$\Rightarrow$ ar. $(ABCD) = \frac{109 + 35}{2}$

$\Rightarrow$ ar. $(ABCD) = \frac{144}{2} = \boxed{72 \text{ sq. units}}$

Q.30 We know that, By Euclid's Algorithm,

$\Rightarrow$ $a = bq + r$

here, $a = 1353$

$b = 861$

$\Rightarrow 1353 = 861 \times 1 + 492$
$\Rightarrow 861 = 492 \times 1 + 369$
$\Rightarrow 492 = 369 \times 1 + 123$
$\Rightarrow 369 = 123 \times 3 + 0$

Now, since remainder $= 0$.
Therefore, b (here, 123) is the HCF.

Ans $\Rightarrow$ HCF $= 123$.

Q.31 To find - Point on X-axis equidistant from points $(5, -2)$ and $(-3, 2)$

Sol, Let point $(+5, -2)$ be A and Point $(-3, 2)$ be B and the ~~Now~~, point on X-axis be P.
Now, WKT - If a point lies on X-axis, then its y-coordinate is 0.
$\therefore$ Co-ordinates of $P = (X, 0)$
Now, According to Question -
$\Rightarrow AP = BP$
$\therefore AP^2 = BP^2$ (Squaring both sides)
$\Rightarrow (X-5)^2 + (0+2)^2 = (X+3)^2 + (0-2)^2$ (By distance formula)
$\Rightarrow (X-5)^2 + \cancel{4} = (X+3)^2 + \cancel{4}$
$\Rightarrow \cancel{X-5 = X+3}$ $X^2 - 10X + 25 = X^2 + 6X + 9$
$[\because (a \pm b)^2 = a^2 + b^2 \pm 2ab]$

$\Rightarrow 6x + 10x = 25 - 9$

$\Rightarrow 16x = 16 \Rightarrow x = \frac{16}{16} = 1$

$\therefore$ Point on x-axis $= (1, 0)$.

Q.32 Taking LHS,

$$\Rightarrow \frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta}$$

$$\Rightarrow \frac{\tan\theta}{1-\frac{1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta} \quad \left[\because \tan\theta = \frac{1}{\cot\theta}\right]$$

$$\Rightarrow \frac{\tan\theta}{\frac{\tan\theta - 1}{\tan\theta}} + \frac{\frac{1}{\tan\theta}}{1-\tan\theta}$$

$$\Rightarrow \frac{\tan^2\theta}{\tan\theta - 1} \cancel{+} - \frac{1}{\tan\theta(-1+\tan\theta)}$$

$$\Rightarrow \frac{\tan^3\theta}{\tan\theta - 1(\tan\theta)} \cancel{+} - \frac{1\ \cancel{\tan\theta}}{\tan\theta(-1+\tan\theta)}$$

$$\Rightarrow \frac{\tan^3\theta \cancel{+} - (1)^3}{\tan\theta(\tan\theta - 1)}$$

$$\left[\because a^3 - b^3 = (a-b)(a^2+b^2+ab)\right]$$

$$\Rightarrow \frac{\cancel{(\tan\theta - 1)}(\tan^2\theta + 1 + \tan\theta)}{\tan\theta\cancel{(\tan\theta - 1)}}$$

$$\Rightarrow \frac{\tan^2\theta + \tan\theta + 1}{\tan\theta}$$

$\Rightarrow \frac{\tan\theta}{\tan\theta} + \frac{\tan^2\theta}{\tan\theta} + \frac{1}{\tan\theta}$

$\Rightarrow 1 + \tan\theta + \cot\theta$ $\left[\frac{1}{\tan\theta} = \cot\theta\right]$

$\Rightarrow 1 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$ $\left[\frac{\sin\theta}{\cos\theta} = \tan\theta \text{ and } \frac{\cos\theta}{\sin\theta} = \cot\theta\right]$

$\Rightarrow 1 + \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$

$\Rightarrow 1 + \frac{1}{\sin\theta\cos\theta}$ $\left[\frac{1}{\sin\theta} = \operatorname{cosec}\theta, \frac{1}{\cos\theta} = \sec\theta\right]$

$\Rightarrow 1 + \sec\theta\operatorname{cosec}\theta$
$= RHS$

Hence, Proved.

Q.34 Given- $p(x) = 2x^3 - \cancel{4}x^2 - 4x + 2$
1^{st} zero $= \sqrt{2}$
2^{nd} zero $= -\sqrt{2}$

To find- Other zero

Sol- Putting value of 1^{st} zero in $p(x)$, we get-

$\Rightarrow 2(\sqrt{2})^3 - (\sqrt{2})^2 - 4(\sqrt{2}) + 2 = 0$

$\Rightarrow 2(2\sqrt{2}) - \cancel{2} - 4\sqrt{2} + \cancel{2} = 0$

$\Rightarrow 4\sqrt{2} - 4\sqrt{2} = 0$

Sol- Since, $\sqrt{2}$ and $-\sqrt{2}$ are zeros
Therefore, $p(x)$ is divisible by

$(x+\sqrt{2})$ and $(x-\sqrt{2})$

Hence, divisible by

$\Rightarrow (x+\sqrt{2})(x-\sqrt{2})$

$\Rightarrow \cancel{x^2-8}\ (x^2-2)$

Diving $p(x)$ by (x^2-2)

$$\Rightarrow x^2-2 \overline{)2x^3-x^2-4x+2} (2x-1$$

$$\begin{array}{r} 2x^3-x^2-4x+2 \\ \cancel{-2x^3} \quad \cancel{+4x} \\ \hline -\cancel{x^2}+\cancel{2} \\ +\cancel{x^2} \cancel{\pm 2} \\ \hline 0 \\ \hline \end{array}$$

$\therefore$ Third factor $= (2x-1)$

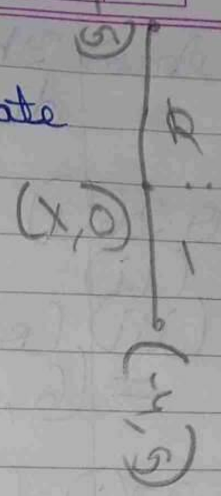
$\therefore$ Third zero $\Rightarrow 2x-1=0$

$\Rightarrow 2x = 1$

$\Rightarrow \boxed{x = \frac{1}{2}}$

Ans $\Rightarrow$ All the zeroes of polynomial $2x^3-4x-x^2+2$ are $\sqrt{2}, -\sqrt{2}$ and $\frac{1}{2}$.

Q.35 Since, the point lies in x-axis, therefore, y-coordinate is 0.
Let the ratio be k : 1
Now, By the formula of internal division



$\Rightarrow y = \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$

$\Rightarrow 0 = \dfrac{k(5) + 1(-5)}{k+1}$

$\Rightarrow 0 = 5k - 5$

$\Rightarrow 5 = 5k$

$\Rightarrow k = \dfrac{5}{5} = 1$

Ans⇒ The ratio of division is 1 : 1.

Since, the ratio is 1 : 1. Therefore, the point is the mid point of line segment.
Now, By Mid-Point Formula,

$\Rightarrow \dfrac{x_1 + x_2}{2} = x$

$\Rightarrow \dfrac{1 - 4}{2} = x$

$\Rightarrow x = \dfrac{-3}{2}$

Ans→ The co-ordinates of point of division is $\left(\frac{-3}{2}, 0\right)$.

Q.36 (i) (1) (12, 0).
(ii) (1) (3, 5).
(iii) Coordinates of Mid Point (S) = (6, 4) Option (3).

Q.37 (i) (3) Parabola.
(ii) (3) 0.
(iii) (3) (−3, 2).

Q.38 (i) (3) 24 cm^2
(ii) (1) RHS.
(iii) (2) 4 : 9

Q.33

$x - y + 1 = 0$		
x	0	−1
y	1	0

$3x + 2y - 12 = 0$		
x	0	4
y	6	0

M T W T F S S

Page No.:

Date:

YOUVA

By plotting these points on graph we get points of triangle as, (2, 3); (4, 0); (−1, 0)

Now, area of this triangle is,

$\Rightarrow ar.(\Delta) = \left| \frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \right|$

$\Rightarrow ar.(\Delta) = \left| \frac{1}{2}[2(0) + 4(-3) - 1(3)] \right|$

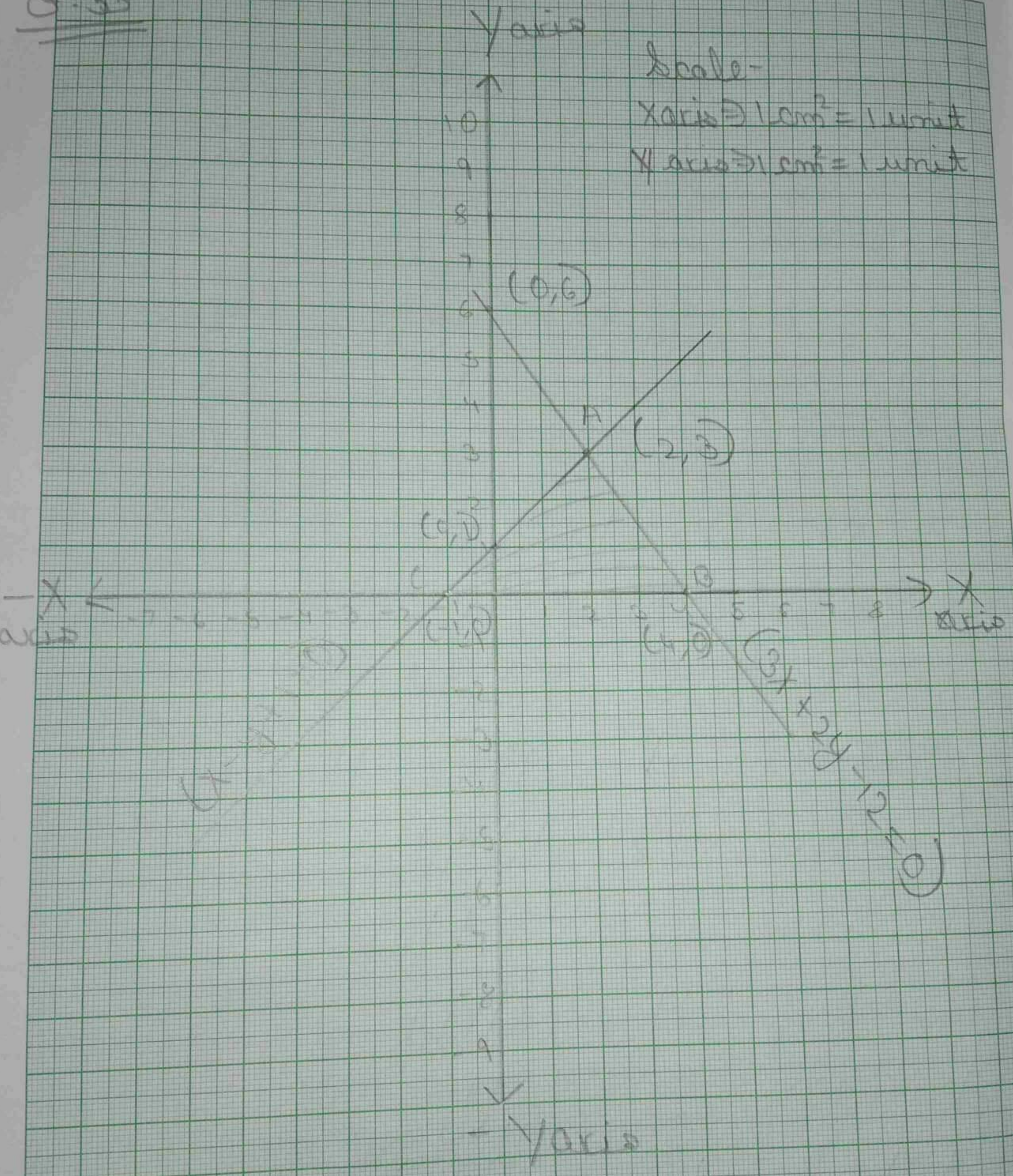
$\Rightarrow ar.(\Delta) = \left| \frac{1}{2}(0 - 12 - 3) \right|$

$\Rightarrow ar.(\Delta) = \left| \frac{1}{2}(-15) \right|$

$\Rightarrow ar.(\Delta) = \frac{15}{2}$ sq. units.

$\Rightarrow ar.(\Delta) = 7.5$ sq. units

Q.33


Y axis
Scale-
X axis ⇒ 1 cm² = 1 unit
Y axis ⇒ 1 cm² = 1 unit
(0,6)
A
(2,3)
(0,1)
C
B
(-1,0)
(4,0)
X axis
-X axis
(3x+2y-12=0)
(x-y+1=0)
-Y axis